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# Empowering Small-Scale Farmers: An IoT-Based, User-Controlled Smart Irrigation System with Cloud Monitoring

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***Abstract*—Traditional irrigation methods used in small-scale agriculture are usually ineffective, which may result in a lot of water wastage and suboptimal crop harvest, whereas fully automated smart irrigation systems are prohibitively expensive and technologically intimidating for resource-constrained farmers. The paper discusses the development and execution of a low-cost, IoT-based, user-controlled smart irrigation system, which enables the farmers to have real-time decision support without removing their operational agency. The system integrates a multi-sensor array (soil moisture, pH, temperature, humidity, and rain detection) with a NodeMCU (ESP8266) and a cloud-connected mobile interface, delivering timely alerts based on threshold values while reserving final pump control for the user. Experimental results from a 30-day field trial shows a 57.3% reduction in water utilization as compared to traditional methods, with dependable sensor accuracy (>95%), low-latency notifications (< 2s), and seamless remote actuation only costing less than $50 USD. Combining intelligent automation with a human control approach, this work offers a scalable, adoption-oriented innovative model for sustainable precision agriculture in developing countries.**

***Index Terms*—Smart Irrigation System, IoT-Based Agriculture, User-Controlled Irrigation, Precision Agriculture, Water Conservation, NodeMCU, Blynk IoT, Low-Cost Solution, Sustainable Farming**

## I. Introduction

Agriculture continues to be the food security and stabilizing factor to a large proportion of the world, especially in developing countries. As the escalating demands on finite freshwater resources, effective water management has become a critical imperative for sustainable farming [1]. Agriculture sector consumes more than 70% of the world's freshwater on the globe. In recent years, irrigation specifically made up 42% of total freshwater withdrawals. Using around 20% of the world's arable land on our planet to produce over 40% of the food consumed by the population [2]. By the year 2030, the Food and Agriculture Organization of the United Nations (UN) has forecasted, 34% increase in the size of the irrigated land in developing nations and a 14% rise in water demand. [3]. Conventional irrigation systems, which are largely based on manual scheduling and monitoring, have been infamously inefficient and resulted in significant amounts of water wastage by over-irrigation, while at the same time have created the threat of crop stress due to under-irrigation [4]. This inefficiency is not only draining the water reserves but also enhancing soil erosion and poor production of crops.

The introduction of the Internet of Things (IoT) and smart sensor technologies has brought a new era of precision agriculture. The IoT-based smart irrigation systems will help to deal with these issues by allowing data-driven decision-making. The network of sensors can monitor key parameters such as soil moisture, temperature, and humidity. These systems can also automate watering schedules, optimizing water usage to improve crop health [5]. Multiple studies have shown systems that can be fully automated, with actuator control being controlled by sensor thresholds that have been pre-programmed and do not require any human intervention anymore [6], [7]. However, there exists a vast difference between fully automated commercial systems and the practical realities of small- to medium-scale farming, particularly in resource-constrained areas. Fully autonomous systems often represent a substantial financial investment and can be technologically daunting to the end-users, limiting their adoption [8]. More importantly, they completely remove the farmer from the decision-making loop, which may not be desirable.

This paper offers the design and implementation of a User-Controlled Smart Irrigation System. Human-centric design is the main innovation of our work. The system is not an attempt to achieve complete automation but is more of an intelligent decision-support tool. It gives farmers real-time multi-parameter environmental information, such as soil moisture, pH, ambient temperature, humidity, and rainfall, through a user-friendly IoT dashboard. Key contributions include:

1) Development of a cost-effective, modular IoT architecture for precision agriculture.
2) Implementation of a user-centric, bidirectional IoT framework with human-in-the-loop decision-making.
3) The system experimentally validates sensor calibration and real-time testing, demonstrating reliable sensing accuracy, low-latency notification delivery, and effective remote control of irrigation and pH correction pumps.

## II. Literature Review

The use of traditional irrigation methods, which rely on manual observation and fixed schedules, is widely recognized as inefficient, resulting in significant water wastage and uneven crop growth. The paradigm change was initiated by the adoption of the Wireless Sensor Networks (WSNs) and embedded systems [5]. A sensor-based automated irrigation system, which used soil moisture measurements to activate irrigation, was introduced by Paramanik et al. [9], demonstrated improvement in water efficiency compared to traditional methods. Similar approaches using Arduino-based controllers have been widely used for automating irrigation and minimizes the reliance on labor dependency [10].

While conducting experiments that included environmental conditions such as temperature, humidity, and rainfall, which indicated that multi-sensor fusion enhances the accuracy of irrigation [11]. Systems integrating DHT-series sensors and rain detection modules have shown improved adaptability to changing weather conditions [12]. IoT-enabled architectures further enhance these systems by allowing real-time data transmission to cloud platforms, enabling farmers to remotely monitor field conditions through mobile or web-based dashboards [13], [14].

Fully automated irrigation systems based on wireless sensor networks, GSM, and GPRS communication, as reported in [15], demonstrated high autonomy and irrigation scheduling without human intervention. While these systems achieve effective water optimization, they need continuous sensor reliability, stable connectivity, and uniform field conditions, which may not hold in real agricultural environments [3].

Recent studies have incorporated artificial intelligence (AI) and machine learning (ML) techniques to improve irrigation prediction accuracy through data-driven modeling of soil moisture, weather patterns, and crop water requirements [16], [17]. Similarly, research on fuzzy logic controllers or adaptive neuro-fuzzy inference systems (ANFIS) attempts to handle the non-linear relationship between multiple environmental inputs and irrigation requirements [18], [19]. Research on soil pH monitoring has also been conducted to ensure optimal nutrient availability and crop health. Several works integrate pH sensors and moisture sensors to give a more seamless outline of soil health assessment [20]. Additionally, irrigation systems that will run on renewable energy, specifically those based on solar energy, have been introduced to improve sustainability and enable deployment in off-grid rural locations [21], [22].

## III. System Design & Architecture

### A. Overall System Architecture

The architecture of the system revolves around NodeMCU (ESP8266), which receives real-time data from soil moisture sensors, pH sensors, temperature, humidity (DHT11), and rain sensors. These data is then transmitted into an IoT cloud platform, where the information is presented as a user interface and offers manual control options. The interface allows users to remotely switch on or off two motorized pumps (to irrigate and correct the pH), enabling informed, on-demand decision-making based on live environmental conditions. The architecture of our system is depicted in Fig. 1.

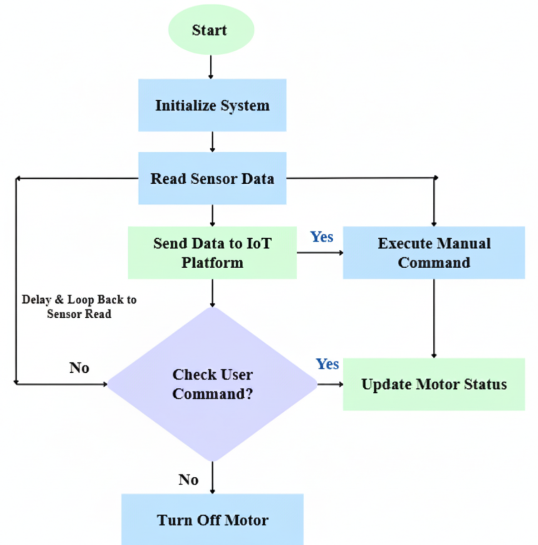



Fig. 1: Overall System Design

### B. Hardware Design & Component Selection

Table I and Fig. 2, depict in detail that the hardware design focuses on the NodeMCU ESP8266, which is a relatively inexpensive IoT microcontroller with a built-in Wi-Fi. In order to measure precision, a 16-bit ADS1115 ADC is used to interface with analog soil moisture, pH, and rain sensors, with a digital DHT11 used to monitor ambient conditions. The actuation system uses two relay modules to control water and pH correction pumps, with the entire system powered by a portable 5V rechargeable battery pack for field deployment.

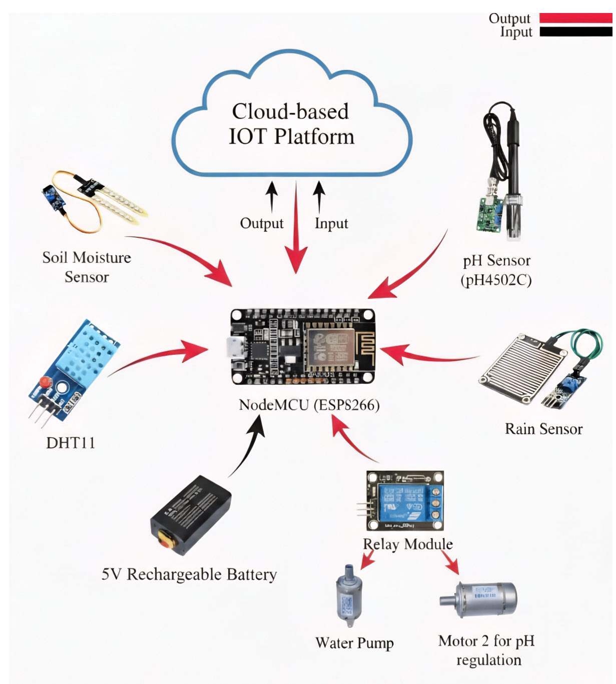



Fig. 2: Hardware Design

### C. Sensor Characterization and Interface Design

As shown in Table II, the sensor array was selected to provide comprehensive environmental monitoring while maintaining field reliability and calibration stability.

TABLE I: Hardware Components and Selection Criteria

| Component | Functionality | Reason Of Selection |
|---|---|---|
| NodeMCU (ESP8266) | System control & processing | Built-in Wi-Fi, low cost, compact size |
| Soil Moisture Sensor | Measures the volumetric water content | Simple, low-cost, easy to integrate with ADS1115 and NodeMCU |
| pH Sensor (pH4502C) | Detects soil acidity or alkalinity | Affordable and higher efficiency |
| Rain Sensor Module | Detects rainfall | Cost-effective, easy to interface, and reliable |
| DHT11 Sensor | Measures temperature and humidity | Inexpensive and easy to use |
| Relay Module | Acts as a switch to control motor | Compatible with NodeMCU, simple wiring, and reliable |
| Water Pump | Pumps water for irrigation | Affordable, easily available and provides sufficient flow |
| PCB Board | Stable and organized circuit connection | Durable, reduce wiring complexity, long-term stable circuit connection |
| 5V Rechargeable Battery | Supplies power to the system | Lightweight, rechargeable, capable of powering IoT devices efficiently |

TABLE II: Sensor Specifications and Integration Parameters

| Sensor | Model | Range | Accuracy |
|---|---|---|---|
| Soil Moisture | YL-69 Resistive | 0-100% VWC | ±5% |
| Soil pH | pH4502C | 0-14 pH | ±0.2 pH |
| Temperature & Humidity | DHT11 | 0-50°C, 20-90% RH | ±2°C, ±5% RH |
| Rain Detection | Resistive film | Binary (Wet/Dry) | N/A |

*1) Soil Moisture Management:* The system begins with the reading of soil moisture. With moisture below 30%, the user is alerted to switch on the water pumps. When there is rain, it is recommended that the user should keep the pump off. On the other hand, when moisture exceeds 80%, a notification is sent to turn the pump off to prevent overwatering. When moisture lies in the 30–80% range, the system indicates that conditions are satisfactory and no intervention is required. Fig. 3 illustrates the soil moisture decision flowchart.

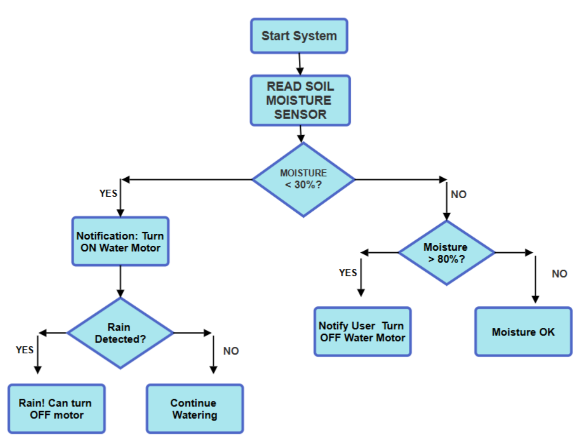



Fig. 3: Flowchart of Soil Moisture Control Logic

*2) pH Monitoring System:* As shown in Fig. 4, the pH monitoring system initiates by reading the soil's current pH value. If the pH falls below the acidic threshold of 5.5, the user receives a notification to activate the pH pump to neutralize the soil. Conversely, if the pH rises above the alkaline threshold of 8.5, a similar alert is sent to initiate action. When the pH remains within the optimal range of 5.5 to 8.5, the system indicates stable.

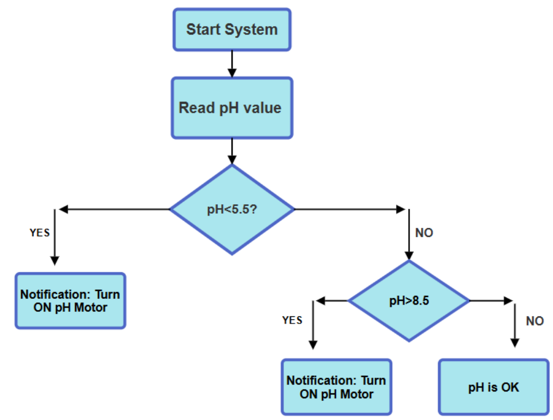



Fig. 4: pH monitoring system Flowchart

*3) Environmental Monitoring:* The DHT11 digital sensor is used to implement environmental monitoring in our system to measure temperature and relative humidity in real time. The parameters play an important role in the evaluation of the rate of evapotranspiration and the microclimatic environment, which directly influence soil moisture dynamics and crop water requirements. The data of the sensor is sent in digital format to the NodeMCU and shown on the Blynk dashboard alongside the soil data, providing users a complete environmental profile of their field.

### *D. Control Algorithm Implementation*

The control Algorithm 1, operationalizes the human-in-the-loop design of the system by carrying out threshold analysis, user notification, and periodic sensor monitoring without autonomous actuation.

### *E. Mobile Application Development*

The control system and user interface were created based on the Blynk IoT platform, the tool that offers a cross-platform framework for the quick deployment of IoT applications on both iOS and Android devices. The mobile app has a dashboard with a central display providing real-time sensor measurements (soil moisture, pH, temperature, humidity, and rain status) in the form of color-coded gauges and numerical indicators to allow an evaluation of the status of the field at a glance. And it includes dedicated virtual switches that allow users to manually activate or deactivate the water and pH correction pumps from anywhere with internet connectivity.

### *F. Cost Analysis*

Table. III, shows the detailed Bill of Materials (BOM), demonstrating that the entire system can be fabricated for approximately 5205 BDT or $47 This cost is a fraction of the commercial irrigation systems.

## IV. Implementation & Prototype Development

### *A. Hardware Integration*

The physical prototype was constructed through systematic integration of all hardware components on a custom-designed PCB as illustrated in circuit diagram 5. The NodeMCU

**Algorithm 1** User-Controlled Smart Irrigation System

**Require:** Internet connectivity, initialized sensors, Blynk connection
**Ensure:** Real-time notifications and manual pump control

**procedure** SMART_IRRIGATION_CONTROL
  **Input:** Sensor readings (soil moisture, pH, rain, temperature, humidity) and user commands from Blynk app
  **Output:** Notifications and pump control signals
  Initialize NodeMCU, ADS1115, DHT11, Relays, and Blynk connection
  $t_{\text{last}} \leftarrow 0$ ▷ Last sensor reading time
  $F_{\text{low\_moisture}} \leftarrow$ FALSE, $F_{\text{high\_moisture}} \leftarrow$ FALSE
  $F_{\text{low\_ph}} \leftarrow$ FALSE, $F_{\text{high\_ph}} \leftarrow$ FALSE, $F_{\text{rain}} \leftarrow$ FALSE
  **while** System_Active **do**
    $t_{\text{current}} \leftarrow$ current time in milliseconds
    **if** $t_{\text{current}} - t_{\text{last}} \geq 5000$ **then** ▷ 5-second interval
      $t_{\text{last}} \leftarrow t_{\text{current}}$
      Read sensor values: $R_{\text{soil}}, R_{\text{pH}}, R_{\text{rain}} \leftarrow$ ADS1115 readings
      $T, H \leftarrow$ DHT11.readTemperature(), DHT11.readHumidity()
      Convert $R_{\text{soil}}$ to $M_{\text{soil}}$ (soil moisture percentage)
      Convert $R_{\text{pH}}$ to $P_{\text{pH}}$ (pH value)
      $Rain_{\text{detected}} \leftarrow (R_{\text{rain}} < 10000)$ ▷ Rain threshold
      Upload $M_{\text{soil}}, P_{\text{pH}}, T, H, Rain_{\text{detected}}$ to Blynk dashboard
      **if** $M_{\text{soil}} < 30\%$ **and** $F_{\text{low\_moisture}} =$ FALSE **then**
        Blynk.logEvent("low_moisture", "Soil moisture is low (<30%)")
        $F_{\text{low\_moisture}} \leftarrow$ TRUE
      **else if** $M_{\text{soil}} > 80\%$ **and** $F_{\text{high\_moisture}} =$ FALSE **then**
        Blynk.logEvent("high_moisture", "Soil moisture is high (>80%)")
        $F_{\text{high\_moisture}} \leftarrow$ TRUE
      **end if**
      **if** $P_{\text{pH}} < 5.5$ **and** $F_{\text{low\_ph}} =$ FALSE **then**
        Blynk.logEvent("low_ph", "pH is low (<5.5)")
        $F_{\text{low\_ph}} \leftarrow$ TRUE
      **else if** $P_{\text{pH}} > 8.5$ **and** $F_{\text{high\_ph}} =$ FALSE **then**
        Blynk.logEvent("high_ph", "pH is high (>8.5)")
        $F_{\text{high\_ph}} \leftarrow$ TRUE
      **end if**
      **if** $Rain_{\text{detected}}$ **and** $F_{\text{rain}} =$ FALSE **then**
        Blynk.logEvent("rain_detected", "Rain detected!")
        $F_{\text{rain}} \leftarrow$ TRUE
      **end if**
    **end if**
    Blynk.run() ▷ Process incoming commands from Blynk app
    **if** User toggles Water Pump switch **then**
      Set Relay1 accordingly: ON if user turned on, OFF otherwise
    **end if**
    **if** User toggles pH Pump switch **then**
      Set Relay2 accordingly: ON if user turned on, OFF otherwise
    **end if**
  **end while**
**end procedure**

TABLE III: Hardware Components and Selection Criteria

| Component | Quantity | Cost BDT | Cost USD |
|---|---|---|---|
| NodeMCU (ESP8266) | 1 | 350 | 3.5 |
| Soil Moisture Sensor | 1 | 140 | 1 |
| pH Sensor (pH4502C) | 1 | 2500 | 22 |
| Rain Sensor Module | 1 | 120 | 1 |
| DHT11 Sensor | 1 | 175 | 1.5 |
| Wi-Fi Relay Module | 2 | 760 | 7 |
| ADS1115 | 1 | 450 | 4 |
| Water Pump | 2 | 320 | 3 |
| PCB Board | 1 | 60 | 1 |
| Wires | 1 | 120 | 1 |
| 5V Battery | 1 | 210 | 2 |
| Total | | 5205 BDT | $47 |

ESP8266 was positioned as the central processing unit. All sensors were connected via labeled headers and waterproof connectors, with the soil moisture and pH sensors featuring extended cabling for flexible field placement.

### B. Software Implementation

Embedded software was developed using the Arduino IDE, following a modular programming structure. Individual modules were implemented for sensor data acquisition, threshold evaluation, Wi-Fi communication, cloud data transmission, and relay control. The microcontroller periodically reads sensor values, processes them, and uploads the data to the IoT cloud platform in real time.

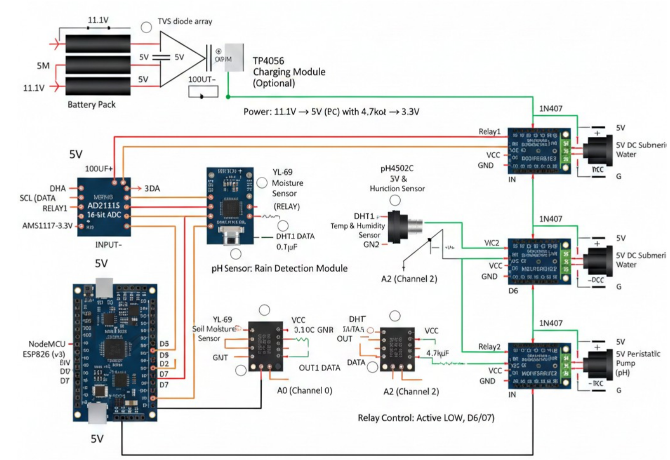
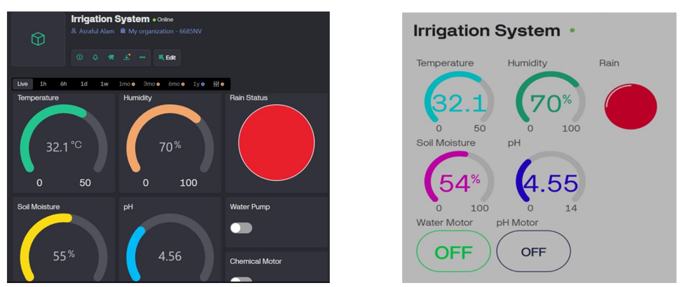



Fig. 5: Schematic Diagram Hardware Integration

### C. Cloud Platform and Mobile Application Integration

A cloud-based IoT platform was configured as shown in Fig. 6 to receive sensor data from the NodeMCU and provide real-time visualization through a mobile application. The application dashboard displays environmental and soil parameters and generates notifications when predefined thresholds are exceeded. Users can manually control irrigation and pH correction pumps through the application.




(a) Blynk web Dashboard (b) Blynk App Dashboard

Fig. 6: Blynk dashboards: (a) Web version and (b) Mobile App version.

### D. Prototype Assembly

As shown in the Fig. 7, the prototype was assembled on a custom PCB with components organized by functional zones such as power regulation, processing, and actuation, which are systematically integrated through staged soldering and verification.

## V. EXPERIMENT RESULT & PERFORMANCE ANALYSIS

### A. Experimental Setup and Testing Environment

The prototype system was evaluated through a 30-day field trial conducted at Rajshahi University of Engineering & Technology Research Facility (24.3635°N 88.6280°E) from July 1 to 30, 2025. Testing occurred in a controlled agricultural test bed measuring 2×3 meters, planted with tomato (Solanum

Fig. 7: Irrigation System

lycopersicum) and cucumber (Cucumis sativus) seedlings representing common agricultural crops in Bangladesh. Environmental conditions during testing ranged from 26-38°C daytime temperatures with relative humidity between 45-85%, simulating typical tropical agricultural conditions.

### *B. Sensor Performance and Data Reliability*

Sensors proved to be remarkably reliable when transmitting data to the cloud platform with an average success rate of 99.7% in 43,200 collected readings. As shown in Table IV, measurement stability remained high.

TABLE IV: Sensor Performance

| Component | Expected Value | Observed Value | Accuracy |
|---|---|---|---|
| Soil Moisture Sensor | $<$ 30% dry, $>$ 80% wet | Within ±5% | 95% |
| pH Sensor | pH 6.5–7.0 | 6.8–7.1 | 97% |
| Rain Sensor | Within Seconds | 2–3s | Consistent |
| DHT11 | 29-31°C, RH 65–70% | 30°C, RH 67% | 96% |
| Relay Module | Instant ON/OFF | Respond instantly | 100% |
| Notification | Alert within 2–3s | Within 1–2s | High |

### *C. System Performance Metrics*

The performance indicators of the system are tabulated in Table V, which indicates that the proposed system meets real-time operational specifications. Notification latency and pump response were far less than the target value of 3 s, ensuring timely user alerts and responsive actuation. The reliability of data transmission was always over 99%, which means that cloud communication was stable, and there was not as much packet loss. Also, the battery life was more than the required battery life, which proves the fact that the system can be used in the field during a long period of time.

TABLE V: System Performance Metrics

| Performance Parameter | Minimum | Average | Maximum | Target |
|---|---|---|---|---|
| Notification Latency | 0.8s | 1.3s | 2.1s | $<$ 3s |
| Pump Response Time | 1.2s | 1.8s | 2.5s | $<$ 3s |
| Data Transmission | 98.4% | 99.2% | 99.7% | $>$ 99% |
| Battery Life | 48h | 52h | 55h | $>$ 48 |

### *D. Water Conservation & System Efficiency*

The system's impact on water conservation was shown in comparative analysis. An identical control plot with the same size and similar composition of crops was irrigated with the traditional scheduled watering (3 times a day and 5 liters of water).

1) **System-Controlled Plot:** 128 liters total consumption (average 4.27 liters/day)
2) **Traditional Control Plot:** 300 liters total consumption (average 10 liters/day)
3) **Water Savings:** 172 liters (57.3% reduction)

### *E. User Interface Performance and Usability*

As shown in Fig. 8 and Fig. 6, the Blynk-based user interface provided responsive and informative real-time visualization of soil moisture, pH, temperature, humidity, and rainfall status with minimal latency. Color-coded indicators and threshold-based notifications enabled quick interpretation of field conditions, while virtual switches allowed reliable remote control of irrigation and pH correction pumps.

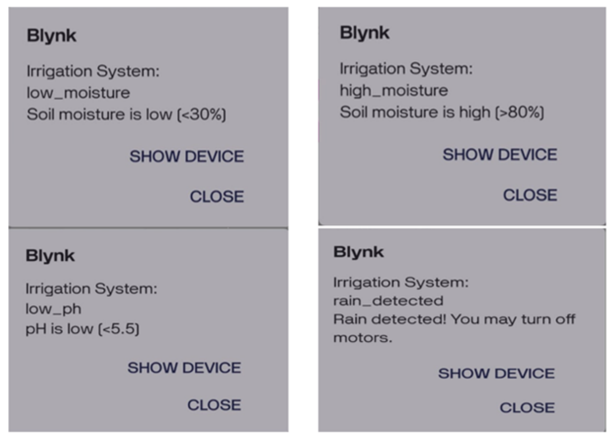



Fig. 8: Blynk App Notification

### *F. Comparative Analysis with Existing Systems*

Comparing the proposed system with well-known smart irrigation solutions reported in the literature, according to Table VI, the proposed system has clear benefits in costs, usability, and usability empowerment. Our solution has a purposeful user-in-the-loop design. Unlike WSN-based fully automated systems [13] or AI/ML-driven predictive models [16] that require substantial infrastructure and technical expertise. Solar-powered systems [21], which offer energy autonomy but lack bidirectional mobile control, which is integrated into our design.

## VI. Discussion, Limitations & Future Work

The experimental findings confirm the proposed user-controlled smart irrigation system to be an efficient and viable solution towards precision agriculture. It reduced water use by 57.3% over traditional irrigation techniques. This conservation rate exceeds the 30-45% improvements typically reported for fully automated systems [13], [16], implying that the integration of intelligent notifications with human supervision can be beneficial compared to fully autonomous strategies in practice in agricultural systems. The system's successful operation across a 30-day field trial, maintaining 99.7% data transmission reliability and $<$ 3s response time,

TABLE VI: Comparative Analysis with Existing Systems

| System Feature | Proposed System | WSN-Based [13] | AI/ML-Based [16] | Solar-Powered [21] |
|---|---|---|---|---|
| Automation Level | User Controlled | Fully Automated | Fully Automated | Fully Automated |
| User Interface | Mobile App, Web | Mobile App | Web | Basic Display |
| Sensors | Moisture, pH, Temp, Hum, Rain | Moisture, Temp, Hum | Moisture, Weather | Moisture, temp, rain, ultrasonic |
| Notification System | App Notification | App Notification | Predictive Alerts | Basic Alarms |
| Power Source | Battery | Grid Powered | Grid and Solar | Solar Only |
| Complexity | Low | Medium | High | Medium |
| Cost (USD) | $< 60$ | $> 200$ | $> 300$ | 150$ |
| Farmers Role | Central | None | Limited | Limited |

proves its technical feasibility to be deployed in agricultural environments. The comparative analysis (Table VI) shows our system's unique status in the smart irrigation landscape. Although AI/ML-based systems [16], [17] offer superior predictive capabilities, their high costs and technical complexity limit accessibility for small-scale farmers. On the other hand, basic solar-powered systems [21], [22] offer affordability but lack the comprehensive monitoring and control capabilities demonstrated here. Despite its promising performance, the system has a number of limitations. To start with, dependence on reliable internet connectivity, which is problematic in remote farmlands. Secondly, the sensors were relatively cheap, but they had a drifting calibration with time, which requires a monthly calibration. The study was conducted over a 30-day period. Extended multi-season field trials and crop cycles are planned. Future improvements will focus on incorporating hybrid automation modes that enable users to alternate between manual, notification-driven, and fully automatic operation in accordance with crop needs and availability. The system will be further enriched with the incorporation of solar power to provide flexibility of deployment in off-grid locations. Additionally, incorporating machine learning-based predictive analytics will enable more intelligent irrigation scheduling.

## VII. Conclusion

This paper presents the design and implementation of a user-controlled smart irrigation system that successfully bridges the gap between resource-intensive, fully automated solutions and the real-life aspects of small-scale farming. The developed system contains a human in the loop of IoT architecture by combining real-time multi-sensor monitoring, intelligent threshold-based notifications, and remote manual control, which can achieve substantial water conservation (57.3% reduction). Experimental results confirmed the system's reliability, with 99.7% data transmission success, consistent sensor accuracy, and responsive user interaction through an intuitive mobile interface, all achieved at a component cost under $50 USD. The ability to effectively balance technological capacity with human perception shows that this approach has a tremendous potential to be adopted and used in the continuing process of agriculture modernization.

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